

support Fm+ low-level output current maximum requirement. Refer to [Section 5.3.13: I/O port characteristics](#) for the I2C I/Os characteristics.

All I2C SDA and SCL I/Os embed an analog filter. Refer to the following table for its characteristics:

Table 61. I2C analog filter characteristics⁽¹⁾

Symbol	Parameter	Min	Max	Unit
t_{AF}	Limiting duration of spikes suppressed by the filter ⁽²⁾	50	260	ns

1. Specified by design – Not tested in production.

2. Spikes shorter than the limiting duration are suppressed.

SPI/I²S characteristics

Unless otherwise specified, the parameters given in [Table 62](#) for SPI are derived from tests performed under the ambient temperature, f_{PCLKx} frequency and supply voltage conditions summarized in [Table 24: General operating conditions](#). The additional general conditions are:

- OSPEEDRy[1:0] set to 11 (output speed)
- capacitive load C = 30 pF
- measurement points at CMOS levels: 0.5 x V_{DD}

Refer to [Section 5.3.13: I/O port characteristics](#) for more details on the input/output alternate function characteristics (NSS, SCK, MOSI, MISO for SPI).

Table 62. SPI characteristics⁽¹⁾

Symbol	Parameter	Conditions	Min	Typ	Max	Unit
f_{SCK} $1/t_{c(SCK)}$	SPI clock frequency	Master mode 2. $V < V_{DD} < 3.6$ V	-	-	24	MHz
		Master transmitter mode 2. $V < V_{DD} < 3.6$ V			24	
		Slave receiver mode			24	
		Slave transmitter mode/full duplex ⁽²⁾ 2.7 V $< V_{DD} < 3.6$ V			24	
		Slave transmitter mode/full duplex ⁽²⁾ 2 V $< V_{DD} < 3.6$ V			22	
$t_{su(NSS)}$	NSS setup time	Slave mode	$4 * T_{PCLK}$	-	-	ns
$t_{h(NSS)}$	NSS hold time		$2 * T_{PCLK}$	-	-	ns
$t_{w(SCKH)}$ $t_{w(SCKL)}$	SCK high and low time	Master mode	$T_{PCLK} - 1$	T_{PCLK}	$T_{PCLK} + 1$	ns
-	SCK low time	Master mode	$T_{PCLK} - 2$	T_{PCLK}	$T_{PCLK} + 2$	ns
$t_{su(MI)}$	Data input setup time	Master mode	4.5	-	-	ns
$t_{su(SI)}$		Slave mode	2	-	-	ns